```
import RPi.GPIO as GPIO
import time

# Pin Definitions
pwmPin = 18
ledPin = 23
butPin = 17
duty = 75
# Setup GPIO
GPIO.setmode(GPIO.BCM)
GPIO.setup(ledPin,GPIO.OUT)
GPIO.setup(pwmPin,GPIO.OUT)
GPIO.setup(butPin,GPIO.IN, pull _ up _ down=GPIO.PUD _ UP)
```

Let's kick it up a notch. Let's add some code for getting a simple button to run with an LED. Update the environment before doing so though.

```
sudo apt-get update
sudo apt-get upgrade
sudo apt-get update
sudo apt-get dist-upgrade
# Display a pretty sine wave in the Python Shell
import math # For the Sine function
import time # For sleep() delay
numCycle = 5 # Number of times we want our sinewave to
cycle
pi = math.pi # Less typing required later = more readable
code

# Function to simplify calling the sine function
def sin(x):
    return math.sin(x)
x = 0
while x < (2 * pi * numCycle):
    bar = int(20*sin(x))     # An integer number for the
length of our bargraph
    x += 0.3                 # the same as saying x = x+0.3
    print ((21+bar)*”=”)     # Print the bar graph
    time.sleep(0.03)
# ‘Breathe’ an LED, change speed with button-press
import RPi.GPIO as GPIO
import math # For the Sine function
import time # For sleep() delay
```